Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Aye Aye Captain
b. Team leader name: Srushti Patil
c. Problem Statement: Sustainability & Social Impact. Older parents living apart from family face loneliness and slow decline that families only notice at a crisis.

Brief about the idea

Saathi is a proactive voice companion for an older adult whose family lives somewhere else.

1 Speaks first

A push says "Saathi wants to talk". When the app opens, Saathi greets the person in their chosen language, using what it already knows.

2 Remembers the family

Caretaker facts, corrections and family stories carry into the next conversation. No fixed daily script.

3 Tells the family what changed

BigQuery compares recent days with the person's own earlier days. The family gets a short, checked note with the parent's own words.

Why it matters Research by others, not results of this prototype. A voice assistant lowered loneliness in a 75+ pilot [5].

347 million

people aged 60+ in India by 2050, up from 149 million in 2022.

[1] UNFPA & IIPS, 2023

5.7% and 9.0%

of people aged 60+, and of women aged 60+, in India live alone.

[2] LASI Wave 1, 2017-18

1 in 6

people are lonely, linked to ~871,000 deaths a year. Isolation and living alone raise early-death risk 26-32%.

[3] WHO, 2025 [4] Holt-Lunstad, 2015

[1] UNFPA and IIPS. India Ageing Report 2023.
[2] IIPS. Longitudinal Ageing Study in India (LASI) Wave 1, 2017-18: India factsheet, persons aged 60 and above.
[3] WHO Commission on Social Connection. From loneliness to social connection: charting a path to healthier societies. 2025.
[4] Holt-Lunstad J et al. Loneliness and social isolation as risk factors for mortality. Perspect Psychol Sci. 2015;10(2):227-237.
[5] Jones VK et al. Reducing loneliness among aging adults: the roles of personal voice assistants. Front Public Health. 2021;9:750736.

Opportunities

How different is it from any of the other existing ideas?

- Software only: no wearables, cameras or room sensors.
- Speaks first and remembers the family between conversations.
- Change is judged against each person's own baseline, not a generic score.
- Culture-aware: a declared religious fast does not trigger a decline alert.
- A natural, human-like companion, not surveillance equipment
- Extends to insurer and employer programmes (see Future Development)

How will it be able to solve the problem?

- Parent: no typing and no forms. Own language, and they can interrupt.
- Family: a Care Desk showing what changed, with the parent's own words.
- Distress is checked on the phone first: one-tap 112 or 108, even if the network drops.
- Negation wins: "main gir nahi gayi" is never logged as a fall.
- The parent has a companion to talk to whenever they want.
- Families get a short note and a human next step, never a diagnosis.

USP of the proposed solution

- Voice-first dignity (zero typing) - elderly speaks naturally in Hindi and Marathi; 8 more languages selectable
- Proactive, not passive - initiates daily check-ins
- Multi-tier Google AI Architecture - on-device check, Gemini Live & Flash, MedGemma (planned)
- Deterministic Safety Guard - never states a diagnosis or a dose
- Longitudinal Drift analysis - BigQuery tracks changes in pauses, speech pace and routine
- Privacy by design - No 24/7 surveillance, closed app never records.

List of features offered by the solution

1 Voice companion

Ten languages selectable; Hindi and Marathi voice-checked. Interruptions handled; quiet mode 9 pm to 5 am.

2 Speaks first

A daily check-in arrives as a push: "Saathi wants to talk". Speech starts when the app is open. A closed app never listens.

3 Memory

Caretaker facts, corrections and family episodes carry into the next greeting. A quiet contact shows an earlier and a later quote.

4 Voice reminders

Created and confirmed by voice. Seen is not confirmed. No dose on the lock screen.

5 Care Desk and Ask Saathi

Family sets reminders, memories and contacts, and gets a daily summary and alerts after each conversation. Ask Saathi updates details by voice or text.

6 On-device safety net

Checks Hindi and English emergency words on the phone (built-in model or keyword list). One-tap 112 or 108, even offline.

7 Drift on own baseline

BigQuery compares the last 3 days with the 11 before on pauses and speaking pace. A declared fast is not read as decline.

8 Distress and nearby help

After a cry for help, Saathi asks what happened. Helplines 112, 108, 14567, nearby places and a Call button.

9 Safety Guard and privacy

No diagnosis, dose or counselling order. Private files, 30-minute links, 7 and 30 day retention.

Process flow diagram or Use-case diagram

A day with Saathi

1

Saathi invites

A push says "Saathi wants to talk". No name, quote or medicine on the lock screen.

→

2

Parent opens

The household is already in the prompt. Gemini Live greets, using a stored memory.

→

3

Parent talks

Reminders, nearby help and the distress check run in the call. Pause and pace are measured; no audio is stored.

→

4

Call ends

One Cloud Task per session. Flash extracts, then embeddings and BigQuery rows follow.

→

5

Family is told

Drift rule and insight, checked by Safety Guard. A quiet line, a notice or an alert.

Special paths

Unanswered or blocked

Snooze counts as a response. Two real unanswered invitations make one notice. A blocked microphone is "device unavailable".

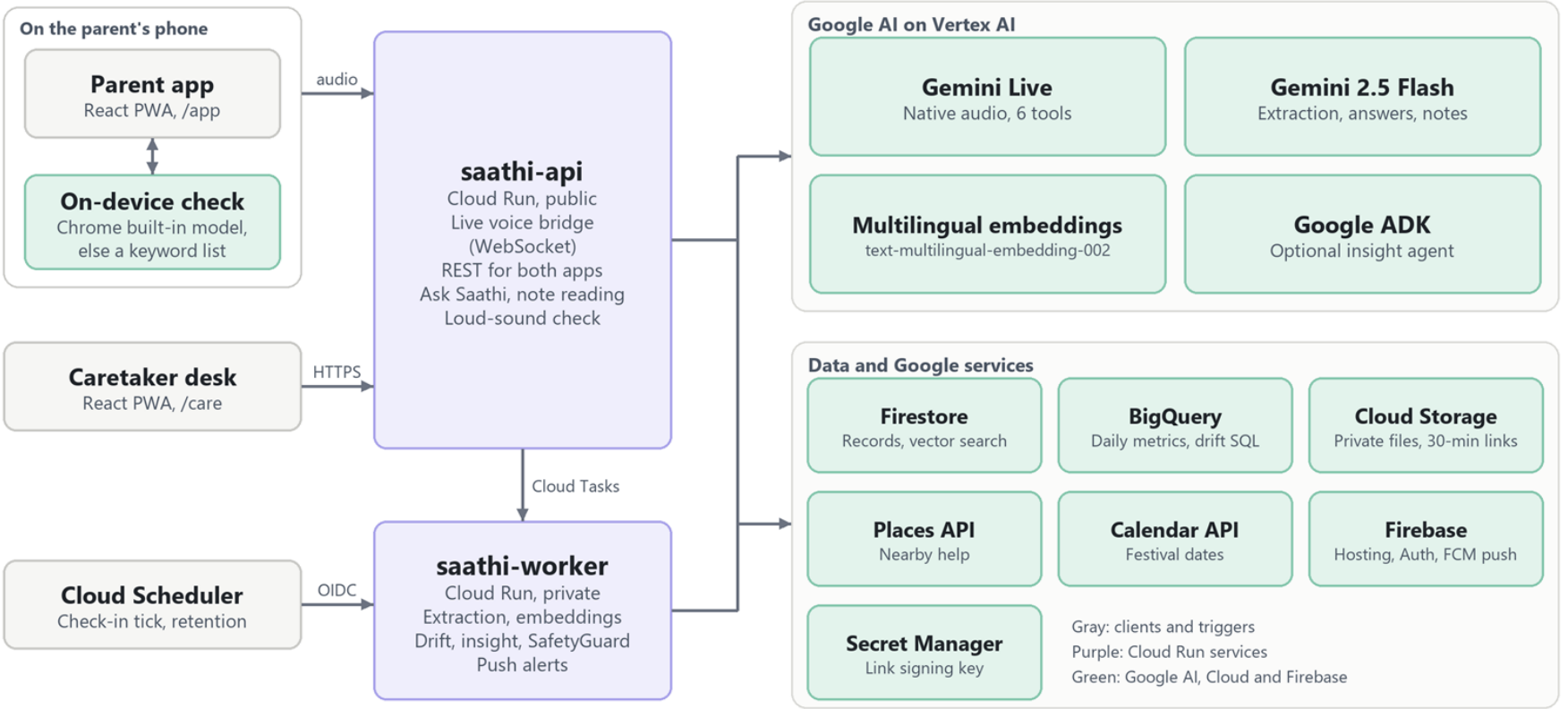
Distress during a call

The phone checks for emergency phrases first and offers one-tap 112 or 108, even offline. Saathi then asks what happened; help or silence tells the family.

Help requested

"Help" or Contact family notifies at once, with a line to call 112. It does not wait for the BigQuery analysis.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Technology	Used for	How we use it well
On-device check (Edge Gemma)	Local distress check, offline 112 / 108 button	Hindi and English words; runs in the browser, no network round trip; the call button does not depend on the cloud
Gemini Live	Real-time Hindi and Marathi voice, barge-in, tool calls	Household preloaded into the prompt; 700 ms silence window so pauses are not cut off; socket opens before Talk
Gemini 2.5 Flash and Google ADK	Post-call extraction, insight, Ask Saathi	Low temperature (0.1); a quote not found in the parent's own words is discarded; ADK tools run in a fixed order
Embeddings and Firestore vector search	Family episodes across time	Household-scoped nearest-neighbour search; no dedicated vector index to pay for
Cloud Run, Cloud Tasks and Scheduler	Public API, private worker, durable jobs, check-in tick	Scale to zero; task names derived from the session id, so a retry cannot run twice
BigQuery	Daily metrics and the drift rule	100 MiB billing cap on every query; one row per household per day
Firebase Hosting, Auth, FCM	Installable app, two roles, push	Generic lock-screen text; role routes the user to /app or /care
Places, Calendar API, Storage, Secret Manager	Nearby help, festival dates, private files, signing key	Field-masked Places calls; holiday calendar read once a day; location blurred to ~100 m; 30-minute links
Application code	API, worker and both web apps	FastAPI, Pydantic; google-genai, google-adk, Cloud client libraries, firebase-admin; React 19, Vite, Redux Toolkit

Estimated implementation cost (optional)

Published prices

- Gemini Live audio: $3 in and $12 out per 1M tokens, 25 tokens per second
- Gemini 2.5 Flash: $0.30 in, $2.50 out per 1M tokens
- Cloud Run (per request): 2M requests, 180k vCPU-s, 360k GiB-s free a month
- Firestore: 50k reads and 20k writes a day free, 1 GiB stored
- BigQuery: first 1 TiB queried a month free, then $6.25 per TiB
- Cloud Tasks: 1M operations free. Scheduler: 3 jobs free

Google Cloud pricing pages, 18 Oct 2026

Per household, per month

$1.75 to $2.85

- About 2 hours of voice: 30 sessions of up to 4 minutes
- Voice is about 90%: audio in about $0.54, audio out $1.1 to $2.2
- Gemini Flash after calls: about $0.10
- Other services stay inside free tiers at this scale
- An estimate from list prices, not a bill

How we keep cost and latency low

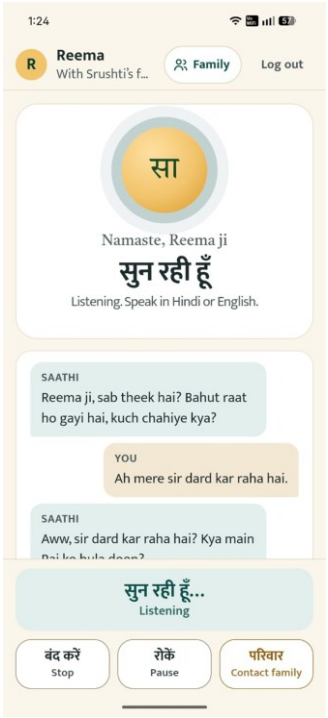
Cost

- Live voice only inside a 240 s session; everything after the call uses Flash
- Prompt built on the server and held to 6,000 characters; sliding-window context compression
- Cloud Run billed per request, zero minimum instances
- Firestore vector search, not a dedicated index; in-process cache protects the free read quota
- BigQuery capped at 100 MiB billed per query

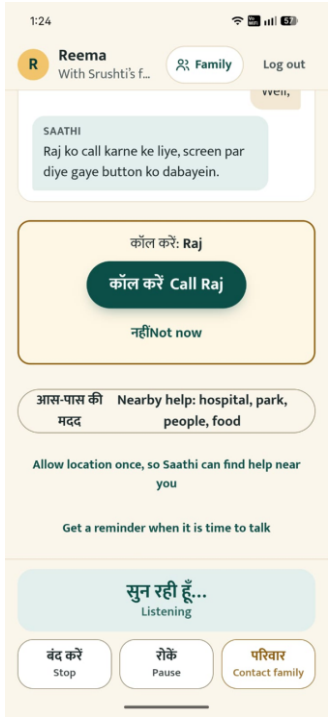
Latency

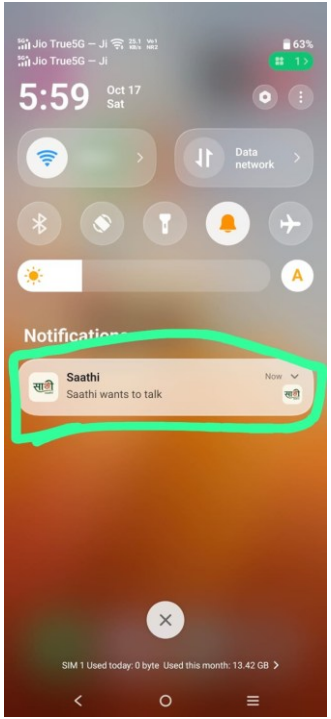
- Socket opens before Talk; household already in the prompt
- Clients pre-warmed at start-up, Cloud Run startup boost
- 40 ms audio frames, gapless playback, instant barge-in
- Distress check on the phone, no network round trip

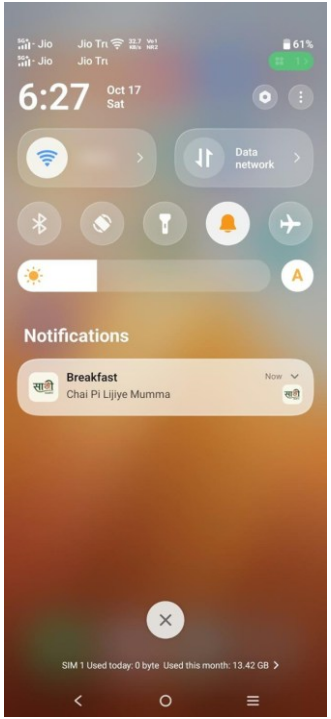
Snapshots of the prototype: the parent's phone





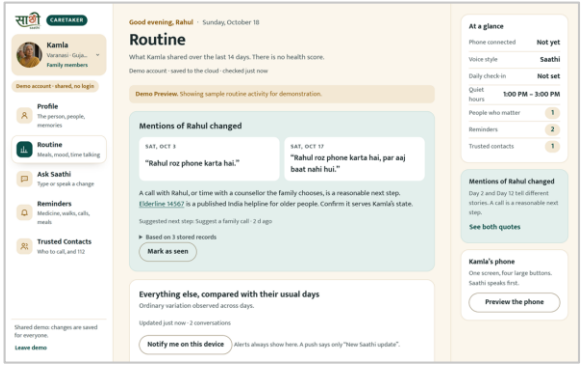


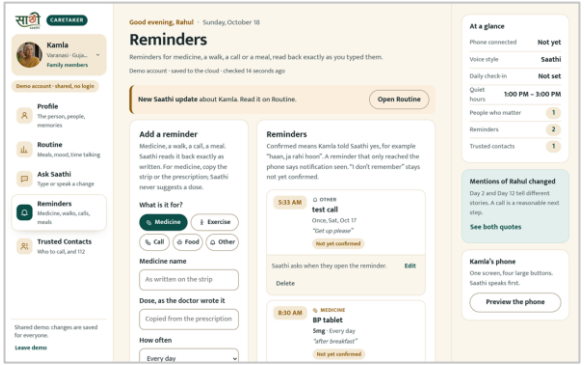


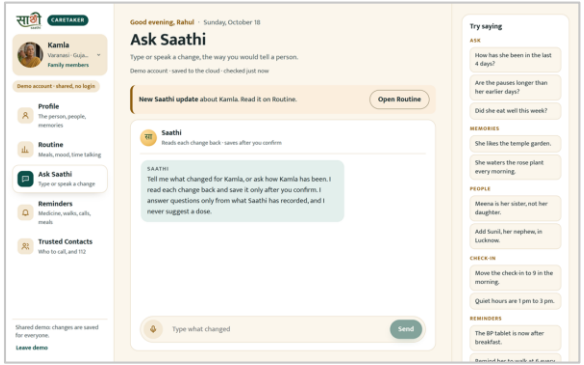


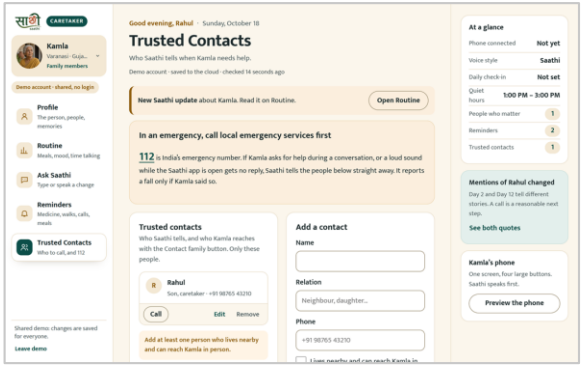
Left to right: Hindi conversation, Marathi screen, one-tap Call button, check-in push, reminder push.
Saathi is a PWA: it opens from a link or installs to the home screen. No app store, and the whole web app is under 2 MB.

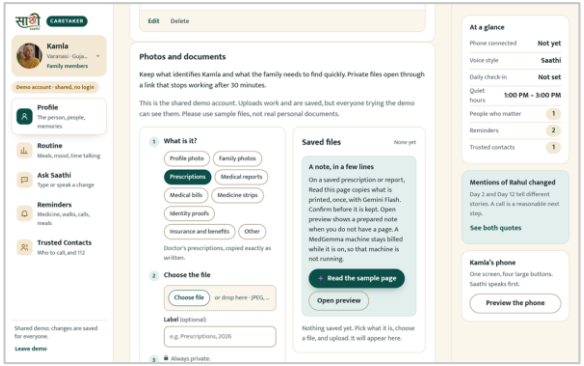
Snapshots of the prototype: the family's Care Desk on a computer

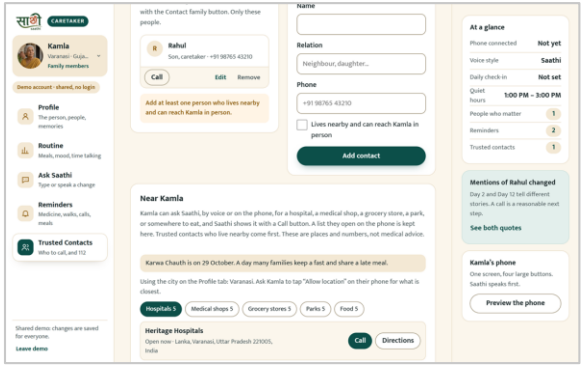






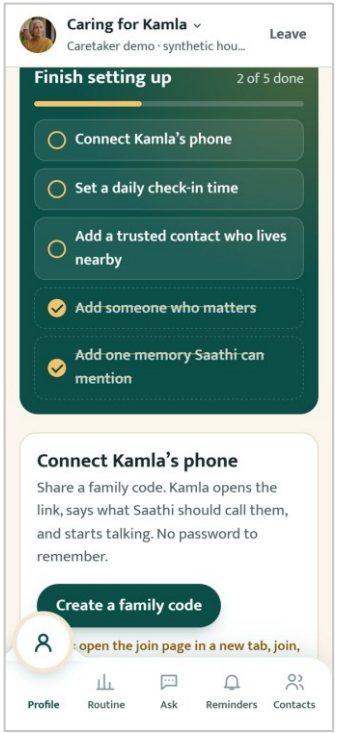


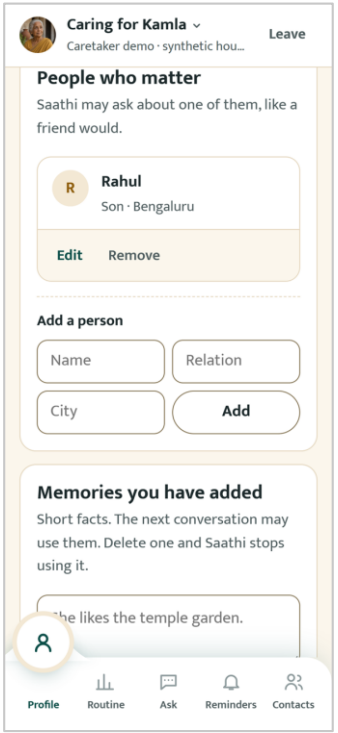


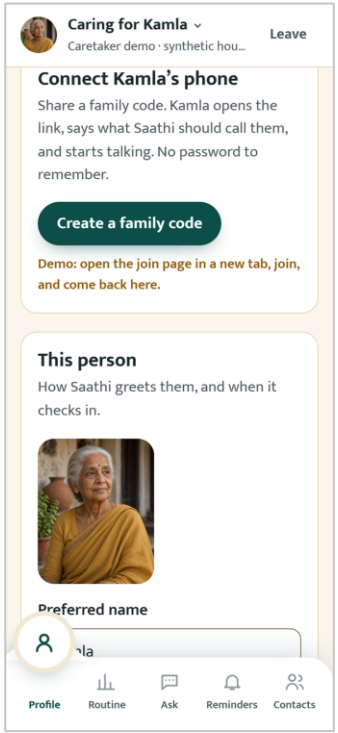


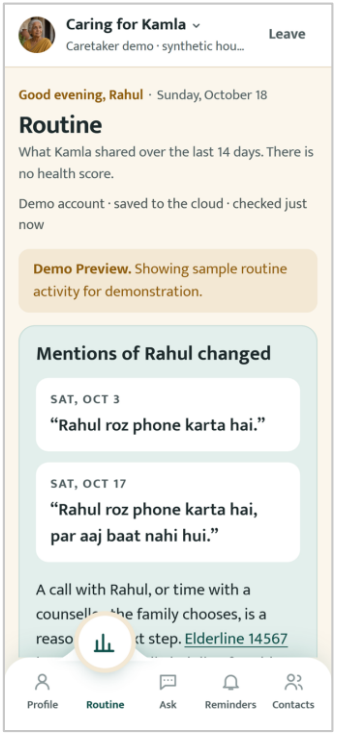
From the live app, synthetic demo household: Routine, Reminders, Ask Saathi, Trusted Contacts, Photos and documents, Near Kamla.

Snapshots of the prototype: the Care Desk on a phone









The same five tabs on a phone. Shown: finishing setup, people and memories, the family code, and how Kamla talks about Rahul.

Prototype Performance report/Benchmarking

Benchmark	Result	How it was measured
Fact extraction	100% (30 of 30)	Roman-script Hindi, Hinglish, English turns in 12 groups: negation, third party, fasting, corrections. Low temperature (0.1); unsupported quotes are discarded.
Drift detection	Flagged on day 3	Synthetic history: 400 ms baseline for 11 days, then 1,100 ms. Flag at z of 2.0 or more against the person's own baseline.
False follow-ups	0 flags	A stable history (500 ms, plus or minus 20) and multi-day fast histories raised no flag.
Safety Guard	25 of 25	18 forbidden sentences blocked (diagnoses, doses, orders, false certainty); 7 safe ones allowed.
Response pause	About 640 ms median	Browser-measured from the parent's last loud frame to the first reply audio; median per session. A sample.
Engineering checks	111 tests, 8 of 8	The 111 API tests pass in under a minute; cloud smoke checks pass (8 of 8, 16 Oct).

Call to insight: about 2.4 seconds end to end



Small steps: Cloud Tasks dispatch 80 ms, OIDC handshake 120 ms, Safety Guard check and Firestore write 40 ms.

Seeded and development runs, synthetic data, approximate times. Not clinical evidence or a service guarantee.

Additional Details/Future Development (if any)

Health insurers and TPA networks

- Problem: decline noticed late can end in an emergency admission.
- Offer: Saathi as a member wellness and early follow-up benefit.
- Pilot hypothesis: earlier, consented family follow-up may avoid some admissions. No percentage is claimed until a pilot measures it.
- Guardrails: member consent, aggregate outcomes only, never used for pricing or claims.

Employers: elder-care benefit

- Problem: staff in Bengaluru, Singapore or Tokyo care for parents from afar.
- Offer: elder-care infrastructure for workforce continuity, as an HR benefit.
- The employee is the caretaker, the parent consents, the employer sees usage only.
- Needs: admin console, single sign-on, multi-household billing.

Product roadmap

- A Nest speaker, after re-checking current Google Home APIs.
- Phone-call and WhatsApp delivery for parents without a smartphone.
- Voice closer to users: an India-region deployment where the Live model is available.
- Consent pause and revoke; Vertex AI Vector Search as the scale path.

Validation before scale

- Pilot with geriatric advisors and an elder-care partner.
- Measure check-in response, time to family follow-up, family-reported peace of mind.
- DPDP review and consent management before real families.
- Real-device tests for push, install, audio and the on-device model; speaker checks for the other languages.

Provide links to your:

1	**GitHub Public Repository**	https://github.com/srushtipatil09/AI_Builder_Cup_2026
2	**Demo Video Link (3 Minutes)**	https://drive.google.com/file/d/1BYQ_IL3YEwDRZ6upRgihbk1uq7Pawfjr/view?usp=drivesdk
3	**Final Product Link**	**https://saathi-511717.web.app** No account needed: try Kamla's phone demo or the caretaker demo from the landing page.

Safety, privacy and responsible AI

No diagnoses

Safety Guard replaces any diagnosis, dose advice or counselling order with a sentence built from stored facts.

A closed app never listens

Check-ins arrive as a push. Speech starts only after the app is open. No wake word, no background microphone.

Honest alerts

A blocked microphone is "device unavailable", never "ignored". A noise is never reported as a confirmed fall.

Data minimisation

Raw audio is not stored. Transcript words are cleared after 7 days, summaries and daily numbers after 30.

Access control

Household and role checks on API routes. The caretaker confirms every change. A parent can join the app with an access code.

Clear boundaries

Synthetic, labelled demo household. Inactivity is simulated, no Nest sensor. Not a medical device or a DPDP or HIPAA certification.

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Build in JAPAC.
Build for the world.

Thank you

